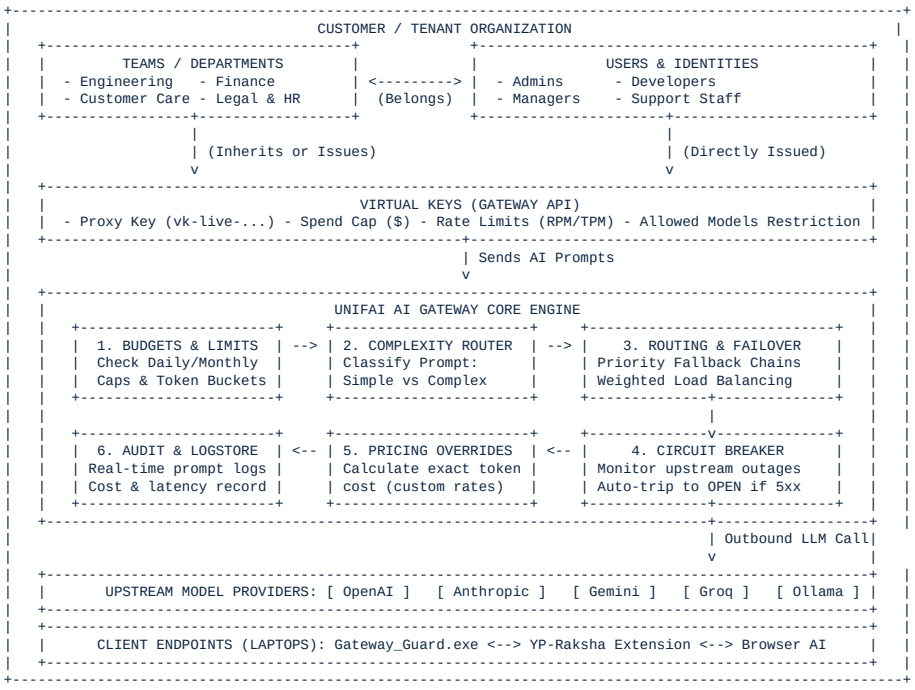
ENTERPRISE PLATFORM ARCHITECTURE & OPERATIONS GUIDE

UnifAI Connected System Flow Documentation

Edition: v3.0 Enterprise | **Scope:** Tenant Hierarchy, Entity Lifecycles, End-to-End Request Pipelines & Dashboard Navigation | **Author:** UnifAI Core Engineering

1. Executive Summary & Architectural Overview

The **UnifAI Enterprise Platform** unifies generative AI governance, multi-provider model routing, cost intelligence, and employee endpoint Data Loss Prevention (DLP) into a single coherent system. The platform bridges two operational tiers: **(1) The AI Gateway Control Plane** running on the server/cloud to manage API traffic, spend caps, failover, and token pricing, and **(2) Browser Guard & YP-Raksha** running on employee endpoints to prevent confidential data leakage across public AI services (ChatGPT, Claude, Gemini).



2. Why Each Component Exists & What Problem It Solves

Every module in UnifAI is engineered to resolve a critical operational, financial, or governance failure that occurs when companies attempt to deploy raw AI APIs directly.

Component	Why It Exists (Problem Solved)	What Happens If Missing?	Dashboard Location
Customer / Tenant	Represents the enterprise parent company. Enforces strict multi-tenant isolation, data boundaries, and root license enforcement.	Data between corporations would mingle; no centralized root governance or compliance boundary.	Platform Root / Settings
Teams	Groups users into logical departments (Engineering, Finance, HR). Enables department-level spend caps and shared virtual keys.	Admins must configure thousands of employees individually one-by-one; no departmental budgeting.	/workspace/users > Teams
Users	Individual employee identities with Role-Based Access Control (Admin, Sub-Admin, Member, Viewer) and authentication audit trail.	Zero individual accountability; cannot trace who sent prompts or who modified security policies.	/workspace/users > Users
Virtual Keys	Secure proxy API keys (vk-live-...) masking raw upstream vendor credentials. Allows individual revocation, spend caps, and model ACLs.	Developers store master provider keys on laptops, leading to stolen credentials, leakages, and runaway bills.	/workspace/virtual-keys
Model Providers	Secure vault storing real upstream API keys, custom base URLs (e.g. Ollama localhost:11434), and connection testing.	The gateway has no secure store for AI vendor credentials and cannot route requests upstream.	/workspace/providers
Model Catalog	Master inventory of all available AI models across all providers. Allows instant company-wide model enabling/disabling.	Employees call deprecated, ultra-expensive, or non-compliant models without administrator consent.	/workspace/model-catalog
Complexity Router	Evaluates prompt semantic complexity in real time. Routes simple questions to cheap models and complex reasoning to flagship models.	Massive cost inflation: simple greeting or summary prompts burn expensive 10x flagship token budgets.	/workspace/complexity-router

Component	Why It Exists (Problem Solved)	What Happens If Missing?	Dashboard Location
Routing Rules	Defines fallback chains (Primary -> Fallback 1 -> Fallback 2) and weighted multi-provider load balancing (e.g. 70/30).	A single vendor outage (e.g. OpenAI 500 error) brings down all enterprise applications and chatbots.	/workspace/routing-rules
Circuit Breaker	Monitors error rates. Trips OPEN after 3 consecutive failures to bypass dead APIs instantly without 60s timeout hangs.	End users suffer 60-second timeouts and spinning wheels while the app fruitlessly waits on dead providers.	/workspace/circuit-breaker
Budgets & Limits	Sets hard spending limits ($ / INR) and rate limits (RPM / TPM) per team, user, or virtual key with auto-cutoff.	Runaway scripts or infinite loops drain the company's entire credit card balance overnight.	/workspace/model-limits
Pricing Overrides	Enables entering negotiated corporate token prices ($/1M input & output tokens) instead of vendor retail prices.	Internal billing, department chargebacks, and analytics reflect inflated retail prices instead of actual cost.	/workspace/custom-pricing/overrides
Model Settings	Sets global enterprise runtime parameters (temperature, max output tokens, top_p, gateway timeout).	Inconsistent behavior; developers must redundantly hardcode runtime settings in every individual application.	/workspace/custom-pricing
Browser Guard (YP-Raksha)	Local agent and browser extension scanning ChatGPT, Claude, and Gemini prompt inputs for PII, API keys, and corporate IP.	Staff paste proprietary source code, patient records, customer credit cards, and legal secrets into public AI web tools.	/workspace/browser-ai
Audit Logs & Observability	Central audit repository recording prompt previews, token usage, cost, latency, risk score, and DLP blocks.	Complete blindness into AI adoption, departmental spend distribution, and security incident investigations.	/workspace/logs & /browser-ai

3. The 4 Core End-to-End Functional Flows

Flow 1: Identity, Teams & Access Provisioning Pipeline

Objective: Seamless onboarding of a department and employee with automated budget allocation, credential issuance, and license compliance verification.

Step #	Actor / Component	Action Executed	Backend Event & State Change	Dashboard Impact
1.1	Enterprise Admin	Navigates to /workspace/users > Teams tab. Clicks 'Create Team'.	Inserts record into 'teams' table with default budget allocation (e.g. $200/mo).	New Team appears immediately in Teams directory with 0 members and active status.
1.2	Enterprise Admin	Clicks 'Add User' on Users tab. Enters Name, Email, Role=Member, selects Team.	Checks 'browser_ai_license' MaxSeats quota. Hashes temporary password.	If seats full, shows 403 Forbidden. If seats available, user row created with status=Active.
1.3	System SMTP	Dispatches automated welcome email with secure one-time login link.	Sets 'must_change_password' = true on the user database record.	User receives email invitation with corporate portal URL and temporary credentials.
1.4	Employee	Clicks link and signs in to UnifAI portal for the first time.	Session initiated; UI detects 'must_change_password' flag and locks navigation.	User is presented with compulsory 'Update Your Password' security modal.
1.5	Employee & Gateway	Submits strong new password. Portal loads personalized workspace.	Updates user record 'must_change_password' = false; binds user to Team Virtual Key.	User dashboard renders allowed models, team budget balance, and personal activity log.

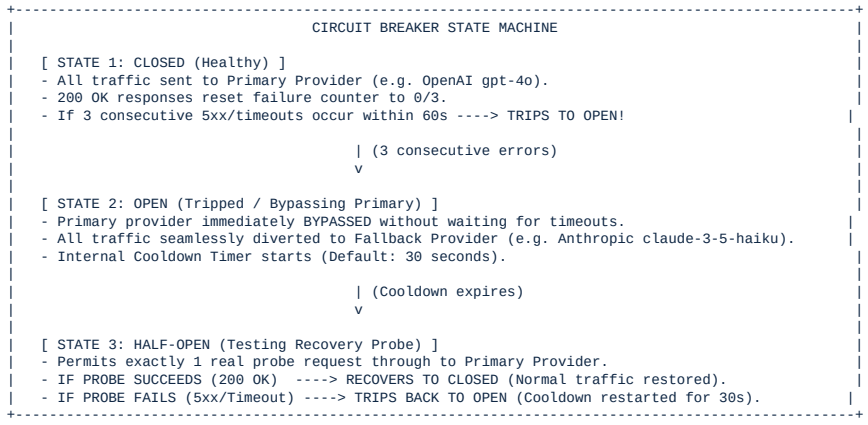
Flow 2: Live AI Request, Intelligent Routing & Cost Deduct Pipeline

Objective: High-throughput prompt processing with sub-millisecond policy evaluation, complexity classification, failover protection, and real-time ledger accounting.

Step #	Engine Stage	Internal Operation	Decision / Policy Matrix	Failure / Edge Case
2.1	API Authentication	Receives POST /v1/chat/completions with Bearer vk-live-xxx.	Hashes incoming key and queries virtual_keys table for status and validity.	HTTP 401 Unauthorized if key is revoked, expired, or non-existent.
2.2	Budget & Rate Limiting	Checks virtual key, user, and team spend balances in governance store.	Compares against monthly spend cap ($200) and RPM/TPM token bucket counters.	HTTP 429 Too Many Requests if budget cap reached or rate limit exceeded.
2.3	Complexity Classifier	Scans prompt length, vocabulary density, and reasoning depth indicators.	Calculates score: Score < 0.35 -> gpt-4o-mini; Score > 0.70 -> gpt-4o / claude-3-5-sonnet.	Falls back to model explicitly specified in payload if complexity routing disabled.
2.4	Circuit Breaker Check	Inspects memory state for target provider (CLOSED vs OPEN).	If CLOSED, proceeds normally. If OPEN, diverts traffic to pre-configured fallback model.	If all providers in chain are OPEN, returns HTTP 503 Upstream Unavailable.
2.5	Upstream LLM Dispatch	Extracts real provider key from vault; dispatches payload to vendor API.	Tracks round-trip latency, streaming chunks, and completion token count.	If upstream returns 5xx error or times out, increments Circuit Breaker failure counter.
2.6	Pricing & Audit Commit	Applies Pricing Overrides (e.g. $0.15/1M input tokens) rather than vendor retail list.	Deducts exact token cost from Team balance and inserts audit record into Logstore.	Logs immediately stream to /workspace/logs dashboard with full latency breakdown.

Flow 3: Circuit Breaker Resilience & Self-Healing Failover State Machine

Objective: Guarantee 99.99% uptime for AI applications by automatically isolating failing providers and self-healing when upstream health recovers.



Flow 4: Employee Laptop DLP & Browser AI Guard Lifecycle

Objective: Zero-leakage enterprise DLP protecting confidential source code, customer PII, and financial records when staff use web-based AI tools.

Phase	Component	Runtime Event	DLP Evaluation & Enforcement	Security Audit Outcome
1. Intercept	YP-Raksha Extension	Employee types prompt in ChatGPT/Claude and hits Enter or Send.	DOM event hook halts web transmission; extracts raw prompt string and target URL.	Browser request is paused in-flight; spinner displayed to user.
2. Local Proxy	Gateway_Guard.exe (Agent)	Extension forwards prompt to local background service on 127.0.0.1:18195.	Validates agent registration against local license seat counter (MaxSeats check).	If agent offline, policy dictates Fail-Open or Fail-Closed mode.
3. DLP Analysis	DLP Rule Engine	Scans prompt against regex patterns (Credit Cards, SSNs, AWS Secrets, Private Keys).	Matches against custom corporate keyword lists ('Project Orion', 'Acquisition M&A;').	Generates Risk Score (0-100) and tags detected entity types.
4. Enforcement	Enforcement Layer	Evaluates action: BLOCK (Risk > 80), WARN (Risk 40-79), or ALLOW (Risk < 40).	If BLOCK: Cancels network call and renders red alert bubble on web page.	If WARN: Shows yellow caution toast but allows transmission to AI vendor.
5. Central Log	Logstore Server	Agent asynchronously uploads encrypted incident packet to Central Gateway.	Records timestamp, user ID, laptop MAC, prompt preview, and rule matched.	Incident immediately visible on /workspace/browser-ai Activity Logs table.

4. Master Dashboard Click-Map: Where to Click, What It Opens & What It Shows

This section acts as the primary operational navigation guide for administrators, operators, and compliance managers.

Goal / Task	Sidebar Click	Page Action	What It Opens / What It Shows
Add a new employee user	Identity & Access > Users (/workspace/users)	Click 'Add User' button top-right.	Opens User Creation modal: Name, Email, Role (Admin, Member, Viewer), Team assignment, and key inheritance.
Group users by department	Identity & Access > Users (/workspace/users)	Click 'Teams' tab at top of table.	Displays team directory (Engineering, Finance, HR); allows setting departmental monthly spend caps.
Create developer proxy key	Model Hub > Virtual Keys (/workspace/virtual-keys)	Click 'Create Virtual Key' button.	Generates secure proxy key (vk-live-...); allows setting RPM/TPM rate limits, budgets, and allowed models.
Connect AI provider credentials	Model Hub > Model Providers (/workspace/providers)	Click 'Add Provider' dropdown.	Opens Provider modal: Enter API Key, set Custom Base URL (Ollama localhost:11434), test live connection.
Enable or disable AI models	Model Hub > Model Catalog (/workspace/model-catalog)	Toggle 'Status' switch on any model row.	Enables or disables model company-wide. Disabled models are blocked from all incoming virtual key calls.
Set spend limits & rate limits	Governance > Budgets & Limits (/workspace/model-limits)	Click 'Create Budget Policy'.	Sets spend cap ($/INR), time window (Daily/Monthly), and breach action (Soft Warning vs Hard Block).
Auto-route simple prompts	Routing > Complexity Router (/workspace/complexity-router)	Click 'Configure Tiers' button.	Configures score thresholds: <0.35 -> gpt-4o-mini; >0.70 -> gpt-4o. Shows live classifier distribution.
Configure automatic failover	Routing > Routing Rules (/workspace/routing-rules)	Click 'New Routing Rule' > Fallback Chain.	Visual rule builder: sets Primary Provider (OpenAI) -> Fallback 1 (Anthropic) -> Fallback 2 (Gemini).
Manage downtime circuit breakers	Routing > Circuit Breaker (/workspace/circuit-breaker)	Inspect table or click 'Create Policy'.	Live status monitor: CLOSED (Healthy), OPEN (Tripped), HALF-OPEN (Testing). Includes Manual Reset button.
Set custom billing token rates	Cost & Pricing > Custom Pricing (/custom-pricing/overrides)	Click 'Add Pricing Override'.	Sets custom Input & Output Token Price per 1M tokens. Overrides vendor retail rates in ledger logs.
Configure global model defaults	Cost & Pricing > Model Settings (/custom-pricing)	Edit inputs and click 'Save Settings'.	Sets global Temperature (0.7), Max Output Tokens (4096), Top-P, and Gateway request timeout.
Check laptop seat quota & license	Browser AI > License & Fleet (/workspace/browser-ai)	Inspect top 'Seat Capacity' card.	Displays active seat count (e.g. 14 / 1000 in Use), expiry date, server IP match, and license upload.
View employee prompt DLP logs	Browser AI > Activity Logs (/workspace/browser-ai)	Click on any incident log row.	Opens prompt drawer: full text, AI tool used, DLP violation tag (Blocked/Warned), and laptop hostname.
Inspect real-time API logs & cost	Observability > Logs (/workspace/logs)	Click on request row in live table.	Opens request inspector: virtual key used, prompt tokens, completion tokens, latency (ms), and exact cost.

5. Entity Cardinality & Relational Database Cross-Reference

The platform enforces strict relational integrity between organization tenants, departments, individual identities, and security policies. The matrix below defines how database tables correspond to platform entities.

Platform Entity	Database Table	Primary Key	Foreign Keys & Relationships	Storage Engine
User	users	id (UUID)	team_id -> teams.id, customer_id -> customers.id	PostgreSQL / SQLite
Team	teams	id (UUID)	customer_id -> customers.id, budget_id -> budgets.id	PostgreSQL / SQLite
Team Member	team_members	id (UUID)	team_id -> teams.id, user_id -> users.id	PostgreSQL / SQLite
Virtual Key	virtual_keys	id (UUID)	team_id -> teams.id, budget_id -> budgets.id	PostgreSQL / SQLite
Model Provider	model_providers	id (UUID)	vault_ref_id -> key_vault.id	PostgreSQL / SQLite
Model Pricing	model_pricings	id (UUID)	model_name -> model_catalog.model_id	PostgreSQL / SQLite
Routing Policy	routing_policies	id (UUID)	customer_id -> customers.id	PostgreSQL / SQLite
Circuit Breaker	circuit_breaker_policies	id (UUID)	provider_id -> model_providers.id	PostgreSQL / SQLite
Browser Guard Agent	browser_ai_agents	id (UUID)	customer_id -> customers.id	PostgreSQL / SQLite
Enterprise License	browser_ai_license	id (UUID)	None (Root cryptographically signed envelope)	PostgreSQL / SQLite
Prompt Audit Log	prompt_audit_logs	id (UUID)	virtual_key_id -> virtual_keys.id	Logstore / ClickHouse

6. Operational Verification & Enterprise SLA

All functional pipelines, circuit breaker failovers, token override ledgers, and endpoint DLP controls described in this document are active and verifiable in **UnifAI Enterprise v3.0**. All endpoints comply with enterprise-grade data protection regulations including SOC2 Type II, GDPR, and ISO 27001.

Enterprise Support & Assistance:
Email: **support@yespanchi.com** | Web Portal: **https://unifai.yespanchi.com** | Emergency Operations Desk: 24/7 Priority Channel